

Win-1 x R1 x

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              10.0.0.1        YES manual up          up
Ethernet0/1              20.0.0.1        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R1#
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R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
```

Win-1 x R1 x

```
R1#
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
S       172.19.0.0/16 [1/0] via 20.0.0.2
S       192.168.168.0/24 [1/0] via 20.0.0.2
R1#
R1#
R1#
R1#
R1#
```

```

Win-1 x R1 x R2 x
R2(config-if)#
R2(config-if)#ip address 20.0.0.2 255.0.0.0
R2(config-if)#
R2(config-if)#no shutdown
R2(config-if)#
R2(config-if)#exit
R2(config)#
R2(config)#interface e0/1
R2(config-if)#
R2(config-if)#ip address 30.0.0.1 255.0.0.0
R2(config-if)#
R2(config-if)#no shutdown
R2(config-if)#
R2(config-if)#exit
R2(config)#
R2(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

R2(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

R2(config)#exit
R2#
R2#show
*Aug 21 13:23:26.430: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    20.0.0.2        YES manual up          up
Ethernet0/1    30.0.0.1        YES manual up          up

```

```

Win-1 x R1 x R2 x
R2(config)#
R2(config)#exit
R2#
R2#
*Aug 21 13:29:34.590: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     20.0.0.0/8 is directly connected, Ethernet0/0
L     20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/1
L     30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
S    172.19.0.0/16 [1/0] via 30.0.0.2
S    192.168.168.0/24 [1/0] via 30.0.0.2
R2#

```

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Win-1 x R1 x R2 x R3 x
R3#
*Aug 21 13:37:24.241: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/2
L    20.0.0.1/32 is directly connected, Ethernet0/2
  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/0
L    30.0.0.2/32 is directly connected, Ethernet0/0
  172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.1/32 is directly connected, Ethernet0/1
R3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    30.0.0.2        YES manual up          up
Ethernet0/1    172.19.0.1      YES manual up          up
Ethernet0/2    20.0.0.1        YES manual up          up
Ethernet0/3    unassigned      YES TFTP  down        down
R3#

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Win-1 x R1 x R2 x R3 x
% Unknown command or computer name, or unable to find computer address
R3#
*Aug 21 13:46:48.010: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
  21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/2
L    21.0.0.1/32 is directly connected, Ethernet0/2
  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/0
L    30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
  172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
R3#

```

```

Win-1 x R1 x R2 x R3 x R4 x
R4(config-if)#
R4(config-if)#no shutdown
R4(config-if)#
R4(config-if)#exit
R4(config)#
R4(config)#interface e0/1
R4(config-if)#
R4(config-if)#ip address 172.19.0.2 255.255.0.0
R4(config-if)#
R4(config-if)#no shutdown
R4(config-if)#
R4(config-if)#exit
R4(config)#
R4(config)#show interface ip brief
      ^
% Invalid input detected at '^' marker.

R4(config)#exit
R4#sh
*Aug 21 13:50:12.267: %SYS-5-CONFIG_I: Configured from console by console
R4#show interface ip brief
      ^
% Invalid input detected at '^' marker.

R4#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    15.0.0.1        YES manual up          up
Ethernet0/1    172.19.0.2      YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down
R4#

```

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Win-1 x R1 x R2 x R3 x R4 x R5 x
R4(config)#
R4(config)#exit
R4#
*Aug 21 13:55:35.620: %SYS-5-CONFIG_I: Configured from console by console
R4#
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
R4#

```

```

Win-1 x R1 x R2 x R3 x R4 x R5 x
R5#
R5#config term
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#interface e0/0
R5(config-if)#
R5(config-if)#ip address 21.0.0.2 255.0.0.0
R5(config-if)#
R5(config-if)#no shutdown
R5(config-if)#
R5(config-if)#exit
R5(config)#
R5(config)#interface e0/1
R5(config-if)#
R5(config-if)#ip address 192.168.168.1 255.255.255.0
R5(config-if)#
R5(config-if)#no shutdown
R5(config-if)#
R5(config-if)#exit
R5(config)#exit
R5#
R5#
*Aug 21 13:58:26.301: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              21.0.0.2        YES manual up          up
Ethernet0/1              192.168.168.1   YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down       down
Ethernet0/3              unassigned      YES TFTP  down       down
R5#

```

```

Win-1 x R1 x R2 x R3 x R4 x R5 x
R5(config)#exit
R5#
R5#
*Aug 21 14:03:59.255: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/0
L     21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
     192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
L     192.168.168.1/32 is directly connected, Ethernet0/1
R5#
R5#

```

```

Win-1 x R1 x R2 x R3 x R4 x R5 x R6 x
R6(config-if)#no shutdown
R6(config-if)#
R6(config-if)#exit
R6(config)#
R6(config)#interface e0/1
R6(config-if)#
R6(config-if)#ip address 192.168.168.2 255.255.255.0
R6(config-if)#
R6(config-if)#no shutdown
R6(config-if)#
R6(config-if)#exit
R6(config)#exit
R6#
R6#
R6#sho
*Aug 21 14:06:29.366: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    162.14.0.1      YES manual up          up
Ethernet0/1    192.168.168.2   YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down
R6#
R6#

```

```

Win-1 x R1 x R2 x R3 x R4 x R5 x R6 x
R6(config)#exit
R6#
R6#
*Aug 21 14:10:16.604: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
     162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     162.14.0.0/16 is directly connected, Ethernet0/0
L     162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
     192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
L     192.168.168.2/32 is directly connected, Ethernet0/1
R6#
R6#

```

